



Evo-Fold: Interactive Protein Stability Platform

An Interactive Computational Pipeline Linking Macromolecular Misfolding Constraints with Genome-Wide Selection Filters



Live Web Application: [LINK](#)



Scientific & Project Background

Genes and proteins rarely act in isolation. During evolutionary timelines, amino acid sequences accumulate mutations, generating genetic diversity. These mutations are subject to the stringent filtering of natural selection.

Inspired by the **Protein Misfolding Avoidance Hypothesis**, **Evo-Fold** bridges dry-lab structural parameters with evolutionary sequence metrics. The core objective is to understand why different genes and structural regions are subject to different evolutionary forces (purifying and positive selection).

Core Research Alignment: Highly expressed proteins face severe selective pressure to fold accurately and avoid toxic cellular misfolding aggregates. This platform evaluates how spatial localization and thermodynamic stability translate into evolutionary conservation.



Key Technical Features

- **Automated PDB Parsing:** Dynamically fetches and parses high-resolution structural coordinates from the RCSB Protein Data Bank using [Bio.PDB](#).
- **Biophysical Integration Stream:** Maps residue-level structural dynamics alongside selective pressure ratios.
- **Interactive 2D Correlation Maps:** Features multi-dimensional scatter maps and sequence distribution streams to easily isolate structural outliers and highly conserved domains.
- **Accessible UI Design:** Engineered completely using an accessible Python setup, removing complex command-line barriers for core biological research.



Analytical Pipeline Workflow

Stage	Process Phase	Key Deliverable / Technical Action

Stage 1	Data Acquisition	Automated fetch of structural coordinate matrices from raw PDB files.
Stage 2	Biophysical Computation	Modeling residue thermodynamic values, stability indicators (ΔG), and spatial parameters.
Stage 3	Evolutionary Processing	Mapping selective intensity via evolutionary metrics (d_N/d_S ratios).
Stage 4	Interactive Visualization	Real-time population of dynamic Plotly charts and sequence distribution graphs.

Software Stack & Dependencies

The entire architecture is modular, leveraging standard industry frameworks:

- **Language Environment:** Python 3.9+
- **Structural Biology Engine:** [Biopython](#) for multi-dimensional coordinate file parsing.
- **Data Science Pipeline:** [Pandas](#) and [NumPy](#) for mathematical arrays and data dataframe transformations.
- **Visualization Layout:** [Plotly Express](#) for generating highly responsive, interactive data streams.
- **Deployment Architecture:** [Streamlit Community Cloud](#) for stable cloud execution.

Local Installation & Execution

Follow these clean terminal steps to setup and test the pipeline environment locally:

1. Repository Setup

Bash

```
git clone https://github.com/YOUR_USERNAME/evo-fold-project.git
cd evo-fold-project
```

2. Dependency Resolution

Bash

```
pip install -r requirements.txt
```

3. Server Initialization

Bash
streamlit run app.py

Future Strategic Roadmap

- Integrate deep learning structural accuracy metrics (like AlphaFold2 pLDDT confidence scores).
- Incorporate real-time solvent accessible surface area (SASA) computing via native DSSP wrappers.
- Implement Machine Learning regression algorithms to predict evolutionary rates directly from geometric 3D coordinate graphs.